SIMPLIS – Manual for ECE 5254 Homework

I. Installing - SIMPLIS ELEMENT

In order to install SIMPLIS Element (Free Version) download the installer from:

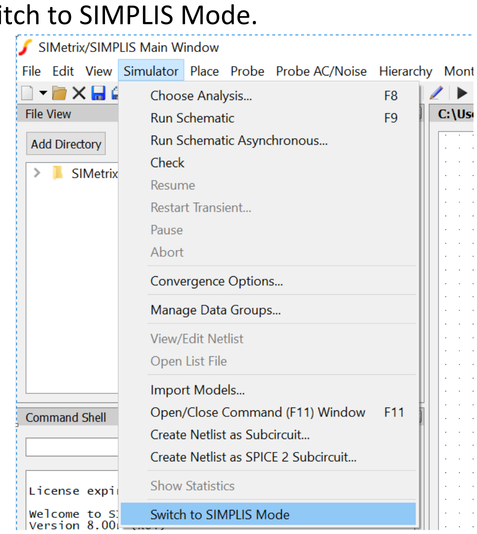
http://simplistechnologies.com/product/elements

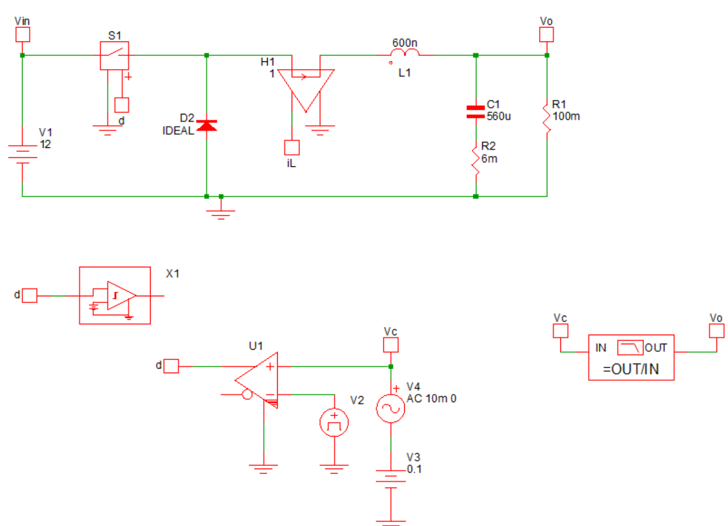
Follow the instruction in the link to install it on your pc.

II. Control to Output Measurement Example

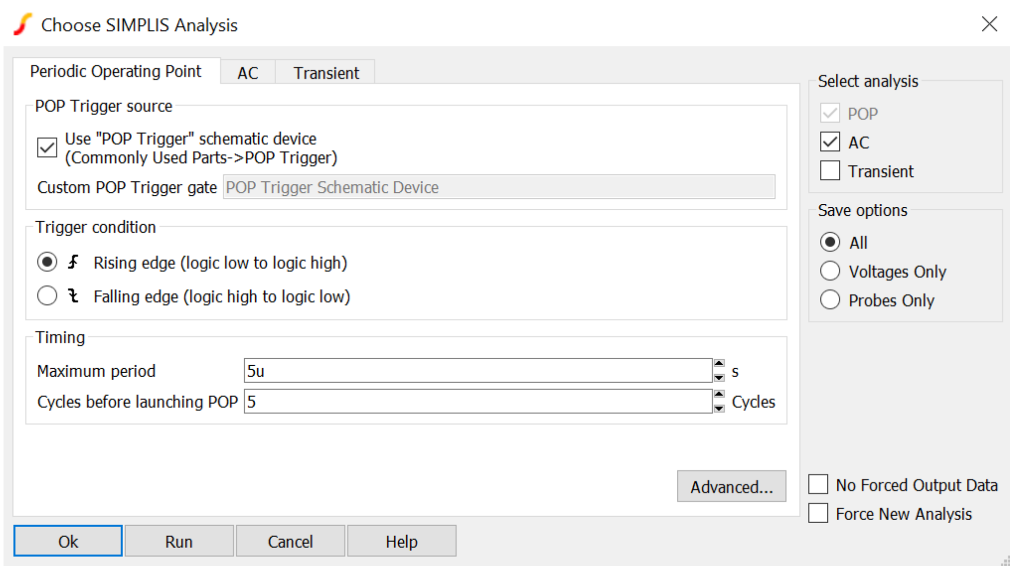
Using the example Buck Open Loop SIMPLIS simulation file, we will do control to output measurement. Follow the step below:

1. Open "Buck_Vc2Vo.sxsch" file.
   Make sure it is opened in SIMPLIS mode. Look at the bottom right corner, it should say SIMPLIS. If not, click Simulator >> Switch to SIMPLIS Mode.

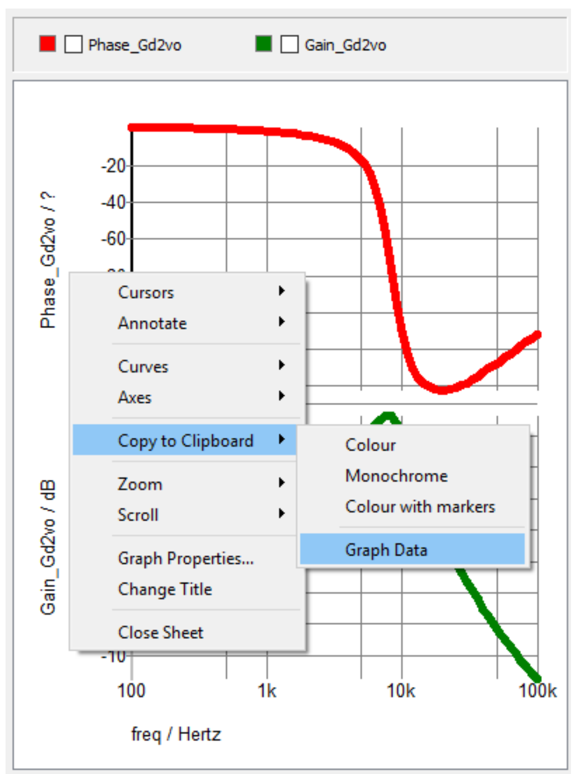




2. Click Simulator >> Choose Analysis, make sure POP and AC are turned on.



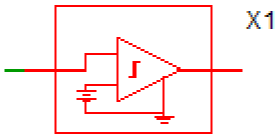
3. Click Run
4. Right Click on the Bode Plot generated by SIMPLIS, choose copy to clipboard >> Graph Data



5. Open a new excel sheet and paste the data on the first column and first row. It is also OK to paste the data in a new text file.
6. Three sets of data are generated: Frequency, Gain and Phase.
7. Save it as “Gm.csv” or “Gm.txt”.
8. You can use MathCAD or MATLab to plot Gm data to compare it with your calculation results

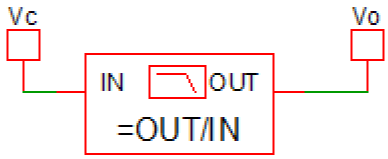
Parts:

1. POP Trigger



In order for SIMPLIS to do AC analysis, we need to ensure the simulation converges in POP mode or Periodic Operating Point. POP trigger works very similarly as Oscilloscope Trigger Function.

2. Bode Plot Generator

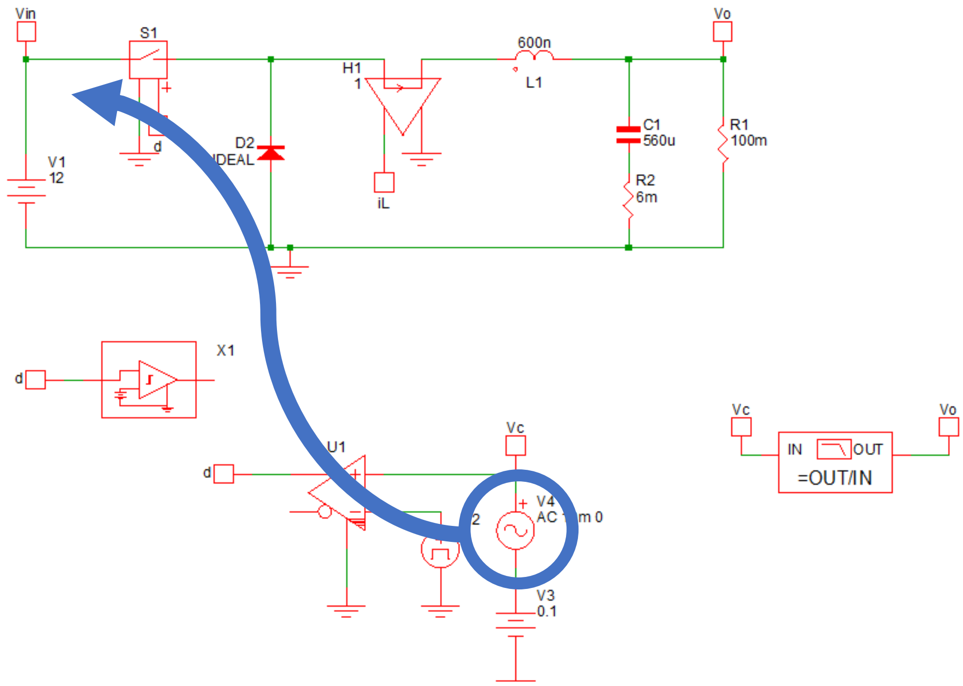


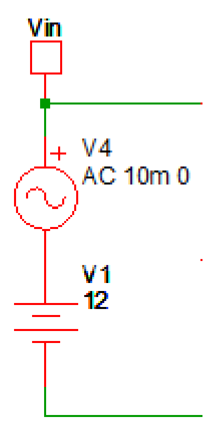
Bode Plot Generator is to generate Bode Plot after POP is done.

III. Input Impedance Measurement Example

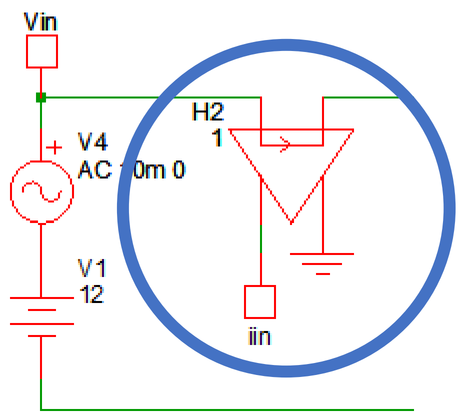
In order to measure input impedance, we have to modulate the input voltage.

1. Move the AC source from control node to the input side of the DC-DC converter in this example of buck converter. Use detach function  on the toolbar.

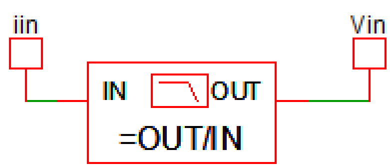




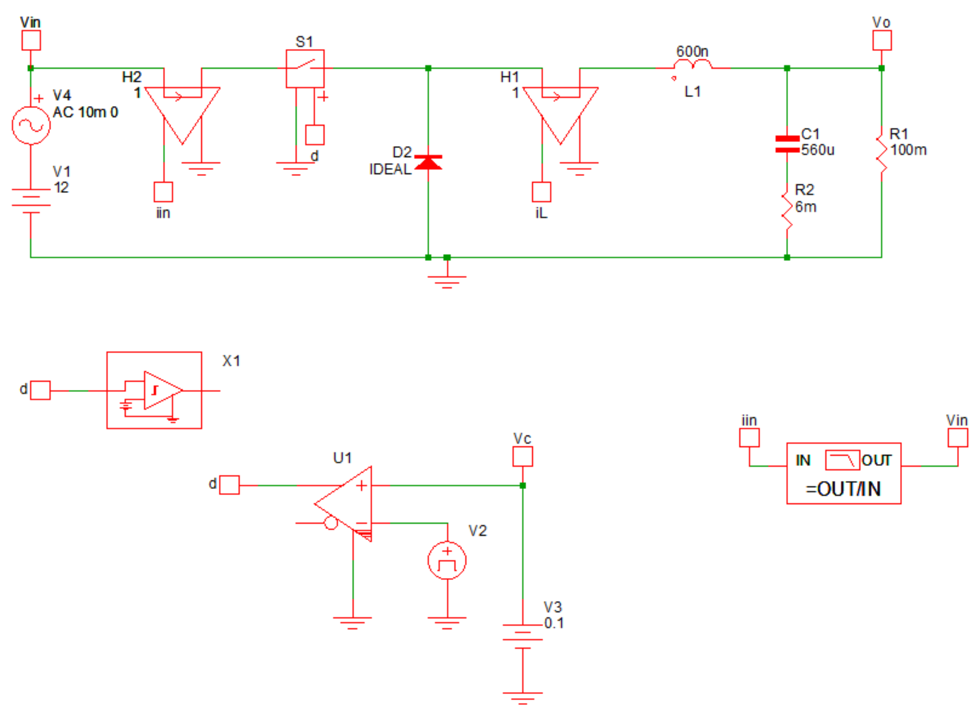
2. Measure the input current by adding a current controlled voltage source, which is similar to the measurement of the inductor current iL.



3. Change the input of bode plot generator to iin instead of Vc. Don't forget to short the disconnected wire on "Vc" when we move the AC source. Also change the output of bode plot generator to Vin instead of Vo.



The modified schematic is as below:

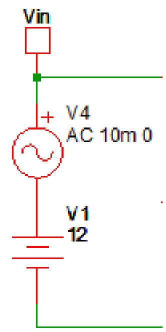


4. Similarly, click Simulator >> Choose Analysis >> Run.
5. Copy the data to clipboard, similar to step 5 and 6 on control to output measurement.
6. Save the data as “Zin.csv” or “Zin.txt”

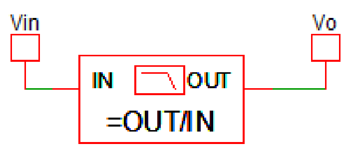
IV. Audio Susceptibility Measurement Example

In order to measure Audio Susceptibility, we also need to modulate the input voltage.

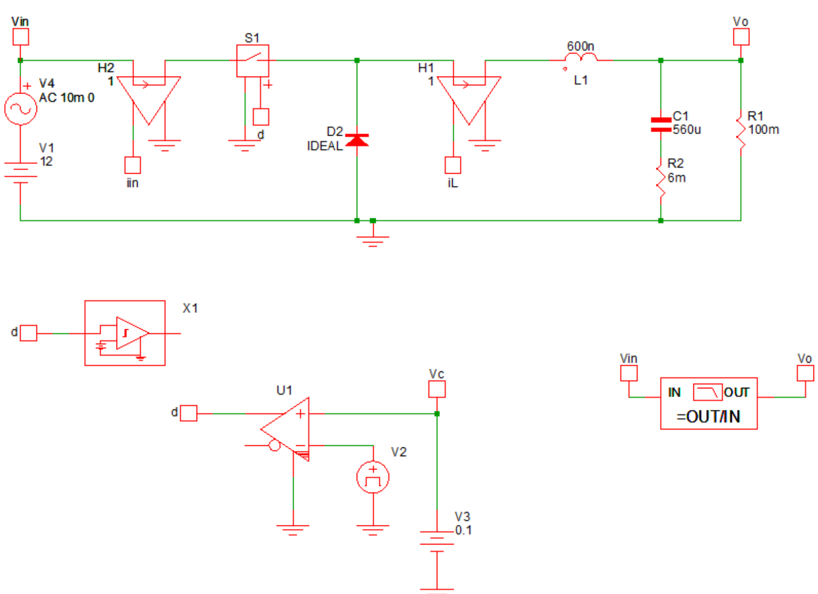
1. Keep the AC Source in series to Input Voltage Vin



2. Change the input of bode plot generator to Vin, and change the output of bode plot generator to Vo.



The modified schematic is as below:



3. Similarly, click Simulator >> Choose Analysis >> Run.
4. Copy the data to clipboard; similar to step 5 and 6 on control to output measurement.
5. Save the data as “Am.csv” or “Am.txt”.